



Smart Horizon 2026: 48-Hour International Hackathon

AIRA: AI Risk & Awareness Assistant

Team Members

1. 1
2. 2
3. 3

Mentor Name

Team Details

- Team ID
- Team Name
- Team Lead
- College Name: PES University RR Campus
- Problem ID:

PROPOSED SOLUTION

Longitudinal Intelligence Engine

- Temporal risk modeling across symptom trajectories — not isolated snapshots
- Detects clinical drift: repeated treatment without improvement signals delayed diagnosis risk
- Adaptive escalation thresholds that evolve with patient history

Pattern-of-Neglect Detection

- Identifies diagnostic delay pathways before they become critical
- Flags patients trapped in ineffective treatment cycles
- Triggers confidence-aware escalation to screening + specialist alert

What Makes AIRA Different

- Decision-support system, not a diagnostic tool — supports clinician judgment
- Offline-first, voice-enabled for low-literacy rural populations
- Explainable outputs: every alert includes temporal reasoning trail and confidence score

TECHNICAL APPROACH

Symptom Capture Layer

- Voice + text input via lightweight mobile app
- Offline-first SQLite storage with temporal metadata; sync-on-connect

03

Adaptive Escalation & Alert System

- Dynamic thresholds calibrated to clinical context and patient risk profile
- Explainable outputs: risk factors, temporal trend visualization, confidence score, and clinical reasoning trail

02

Temporal Risk Scoring Engine

- Transformer-based NLP for symptom classification and semantic clustering
- Hybrid ML + rule-based reasoning: longitudinal model detects clinical drift patterns, treatment response lag, and escalation triggers

04

Clinician Decision Support Dashboard

- Web portal with prioritized patient queue ranked by escalation confidence
- Stack: Python, FastAPI, React, TensorFlow Lite, PostgreSQL

FEASIBILITY

Why Buildable in 48 Hours

- Prototype scope: temporal symptom tracking + clinical drift detection + adaptive escalation
- Pre-trained NLP models (Hugging Face) reduce development time
- Modular architecture: each layer independently testable and deployable
- Synthetic longitudinal datasets available from open cancer registries

Challenges & Mitigations

Challenge: Limited longitudinal patient data

Mitigation: Synthetic temporal data generation + ICMR registry datasets

Challenge: False positive escalation in low-risk populations

Mitigation: Multi-threshold validation + clinician override with confidence scoring

Challenge: Rural connectivity and offline reliability

Mitigation: Offline-first PWA with background sync and local caching

IMPACT ON SOCIETY

-  **Diagnostic Delay Prevention** Detects clinical drift patterns before diagnosis failure occurs. Acts as a safety net in primary care, catching patients stuck in ineffective treatment loops and reducing diagnostic delay pathways that lead to late-stage presentation.
-  **Rural & Urban Equity** Voice-first, low-bandwidth design reaches underserved populations. It bridges the diagnostic gap in primary care across the urban-rural divide and empowers patients with a documented symptom history for self-advocacy.
-  **Clinical & Human Impact** Prevents late-stage diagnoses caused by symptom dismissal, improves screening adherence through intelligent escalation, and restores patient agency in their own health journey.
-  **SDG 3 Alignment** Supports SDG 3.4 by helping reduce premature mortality from non-communicable diseases, and SDG 3.8 through universal health coverage with intelligent triage.

RESEARCH AND REFERENCES



World Health Organization (WHO)

- WHO Cancer Early Detection Guidelines, 2023
- Global Cancer Observatory (GCO) — delayed diagnosis data



Indian Council of Medical Research (ICMR)

- National Cancer Registry Programme (NCRP) Reports
- ICMR Population-Based Cancer Registry data



Peer-Reviewed Studies

Delayed cancer diagnosis — *The Lancet Oncology*, 2022

AI symptom checker accuracy — *Nature Medicine*, 2023

Misdiagnosis patterns in primary care — *BMJ Quality & Safety*



Screening Guidelines

- NCCN Cancer Screening Guidelines v2.2024
- USPSTF Cancer Screening Recommendations

Thank You



"Because early signs should not be ignored twice."

Real-World Scenario



A 42-year-old woman in rural Karnataka reports persistent abdominal pain over 3 months. Her local clinic prescribes antacids twice — no improvement. AIRA detects the clinical drift pattern, escalates her risk score, and alerts a district oncologist. She is screened within 2 weeks. Early-stage cancer is detected. Treatment begins immediately. Her life is saved by a system that recognized what was missed.

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